

In Silico Identification of Novel Compounds for Insomnia Disease

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1. Introduction

Insomnia is a prevalent sleep disorder which is clinically characterized as patients who present difficulty falling asleep or staying asleep and daily functioning impairments; this difficulty must be maintained in time from up to a week within a three-month period to meet clinical criteria [Naha et al., 2024].

Insomnia disease has been linked with multiple medical conditions, both originating them or worsening the symptoms related to such conditions. These conditions include heart disease, neurological diseases, chronic urinary and gastrointestinal disorders [Taylor et al., 2007]. In addition, psychiatric diseases are the most prevalent among insomnia disease patients, as up to 40% of the patients are diagnosed at some point of their life with disorders such as depression, anxiety and post-traumatic stress disorder among others [Mai and Buysse, 2008]. Regarding the treatment of insomnia disease, it includes both non-pharmacologic and pharmacologic interventions. The so-called non-pharmacologic interventions such as Cognitive Behavioral Therapy (CBT) for insomnia are tried before pharmacologic intervention as an attempt to minimize the risk of medication side-effects; however, in approximately 20% of the cases this CBT therapy does not work and pharmacologic-based interventions are needed.

Regarding the wide range of pharmacotherapeutic agents that can be used, the most frequently prescribed drugs are benzodiazepines as positive allosteric modulators of the GABA receptor complex, presenting a very fast action and being well tolerated. However, it is key to highlight that they are meant to be used for short periods of time, as they are highly addictive and present severe side-effects, including cognitive deterioration and tolerance. They affect “sleeping architecture” [Álamo and López-Muñoz, 2010], producing anxiety and withdrawal syndrome if they are abruptly discontinued [Sateia et al., 2017]. On the other hand, melatonin receptor-targeted drugs are excellent for short periods of time but present huge limitations in order to treat chronic insomnia [Ferracioli-Oda et al., 2013].

Under the current panorama, the rise of a new therapeutic approach is based on targeting and inhibiting the Orexin Receptor 2 (OX2), which is part of the so-called “Orexinergic System” — a neurophysiological system tightly linked with sleep-wake cycles. Orexin receptor antagonists target these receptors in order to block the wake phase of the cycle, producing a more natural transition to sleep. It has been proved that drugs approved for this target preserve sleep phases (REM and NREM), do not affect sleep architecture, do not produce addiction nor consequent withdrawal syndrome. For all these reasons, orexin receptors are a desirable target to develop new drugs for insomnia disease treatment.

This work aims to use an *in silico* screening workflow in order to identify novel molecules capable of modulating the human Orexin 2 receptor, a key regulator of the sleep-wake cycle. This research integrates three sequential screening methods: a QSAR method, a pharmacophore model, and molecular docking as the pivotal validation step. Beyond the novel identification of compounds, this work aims to evaluate the reliability of the computational workflow by integrating appropriate controls and evaluating the identified compounds as safer alternatives to current drug approaches.

2. Materials and Methods

Firstly, a search was conducted on the Protein Data Bank (PDB) in order to retrieve the experimental structure of the chosen target, the human Orexin 2 receptor, obtaining as the best entry the accession (UniProt ID: O43614, PDB entry: 4S0V). This structure was obtained by X-ray crystallography by Yin et al. (2015) at a resolution of 2.50 Å and is bound to Suvorexant, a well-known reference ligand for OX2 receptors. Since this is a widely studied ligand, the binding pocket presents a conformation ready to bind similar ligands; this approach fits within the induced-fit framework, as the receptor architecture is already stabilized in a high-affinity binding state, reducing uncertainty and allowing better predictive accuracy during virtual screening.

Regarding the chosen screening library, the Enamine GPCR-Targeted Library was selected, as the histamine OX2 receptor belongs to the G protein-coupled receptor (GPCR) superfamily. Compounds from this library underwent multiple filtering steps to ensure drug-likeness and lead-likeness. For physicochemical filtering, Lipinski’s Rule of Five was applied to maintain bioavailability. Following this first filter, and given the target’s location in the Central Nervous System (CNS), a more strict filter was implemented based on Pajouhesh & Lenz (2005) guidelines. Lastly, a Quantitative Estimate of Drug-Likeness (QED, RDKit descriptor; [Bickerton et al. 2012](#)) threshold of $QED > 0.6$ was established to prioritize molecules with favorable pharmacokinetic characteristics. Filtering for unwanted substructures and Pan-Assay Interference Compounds (PAINS) was then conducted to avoid false positives in the virtual screening [[Brenk et al., 2008](#)].

To optimize the computational cost of the following screening steps, the library was clustered and reduced through Butina Clustering [Sydow et al., 2022], using Morgan Fingerprints and a Tanimoto similarity threshold. The library was grouped into clusters of chemically related families, and the centroids of each cluster were selected to ensure the final screening represented the maximum chemical diversity of the Enamine library.

A Quantitative Structure-Activity Relationship (QSAR) model was developed using a training set of known OX2 antagonists retrieved from ChEMBL (UniProt: O43614) [Zdrazil et al., 2024]. A Random Forest regressor was trained on molecular fingerprints and 2D/3D molecular descriptors [Pedregosa et al., 2011]. The model was validated using known drugs (Suvorexant and Lemborexant) as positive controls, achieving high predictive accuracy. Compounds with the highest predicted pIC_{50} values advanced into subsequent steps.

To complement the ligand-based QSAR method with structure-based information, a pharmacophore model was generated. This model aims to capture the essential interactions between FDA-approved drugs and the 4S0V (OX2) binding pocket. The pharmacophore model was validated using positive controls from OX2 receptor antagonists (SB-649868 and ACT-462206) and negative controls using benzodiazepines that present sedative effects but do not target OX2 receptors (Lorazepam and Zolpidem).

The final stage of the virtual screening pipeline involved molecular docking to evaluate the binding mode and thermodynamic affinity of the 5 top candidates against the human Orexin 2 receptor (OX2). The crystal structure (PDB ID: 4S0V) was prepared by removing solvent molecules and the co-crystallized ligand (Suvorexant); hydrogen atoms were added to satisfy valency and Gasteiger charges were assigned. To validate the predictive reliability of the docking parameters, a redocking procedure was performed and evaluated through Root Mean Square Deviation (RMSD).

Molecular docking was computed using AutoDock Vina [Eberhardt et al., 2021, Trott and Olson, 2010] within UCSF Chimera [Pettersen et al., 2004]. The search space (grid box) was centered on the binding site targeting residue Asn324 with coordinates (58.856, 7.542, 55.199) Å and box dimensions of $25 \times 25 \times 25$ Å. The resulting docking poses were ranked based on their binding affinity scores (kcal/mol). Visual inspection was performed to identify hydrogen bonds using the "FindHBond" function, steric complementarity using "Clashes and Contacts," and 2D interaction maps to visualize hydrophobic arcs and π -stacking interactions. Several compounds were used as positive and negative controls to validate the docking methodology.

3. Results

The initial screening began with the Enamine GPCR-Targeted Library, which included 53,440 compounds. After applying the previously described filters, the library was significantly reduced to 43,849 compounds, from which 1,259 were classified as PAINS or undesirable structures. Using the proposed clustering analysis, the filtered library was grouped into clusters representing distant chemical families; selecting the centroids of these clusters allowed for a less computationally demanding analysis while maintaining exploration of the full chemical space.

The Random Forest QSAR model demonstrated high reliability. To validate its predictive power, FDA-approved drugs were used as positive controls, obtaining predicted pIC_{50} values of 7.39 and 7.48 for Suvorexant and Lemborexant, respectively — the two most widely used OX2 receptor-targeting drugs for insomnia treatment.

The pharmacophore model served as the last filter before docking. Negative controls (Lorazepam and Zolpidem) presented very low pharmacophoric scores, while targeted OX2 receptor antagonists presented high scores, confirming model specificity (Figure 1).

	Name	Score	Match_Percent	Control
1	ACT462206	9	69.230769	Positive
0	SB-649868	7	53.846154	Positive
2	Lorazepam	4	30.769231	Negative
3	Zolpidem	2	15.384615	Negative

Figure 1: Pharmacophore validation results. Positive controls (OX2 antagonists) score highly, while benzodiazepine negative controls score poorly.

After pharmacophore scoring, the best five compounds advanced into docking analysis: Z1155102784, Z1840003394, Z52198758, Z28143004, and Z26441190 (Figure 2).

Molecular docking was the pivotal validation step. The redocking of Suvorexant into the 4S0V pocket yielded an RMSD of 0.22 Å (Figure 3), confirming the reliability of the AutoDock Vina scoring function for this receptor.

The top-scored compound, Z26441190, achieved a binding affinity of -10.49 kcal/mol. Visual inspection of the binding site revealed that the compound fits precisely within the

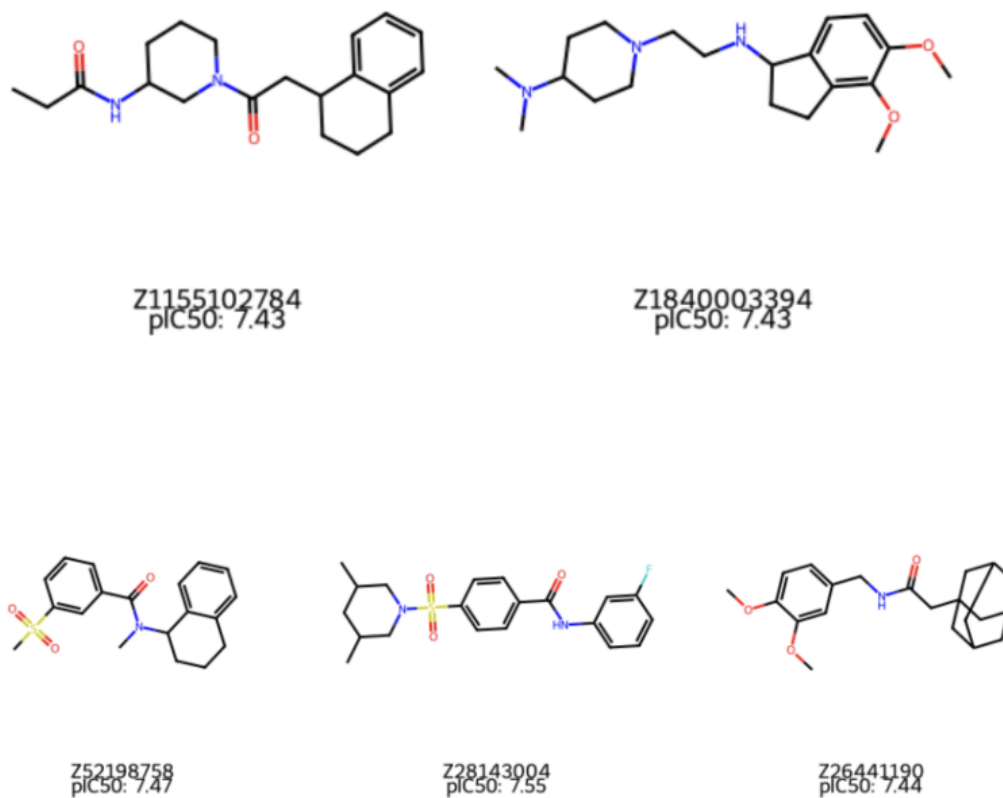


Figure 2: Molecular representation of the five best-scored compounds after QSAR and pharmacophore filtering.

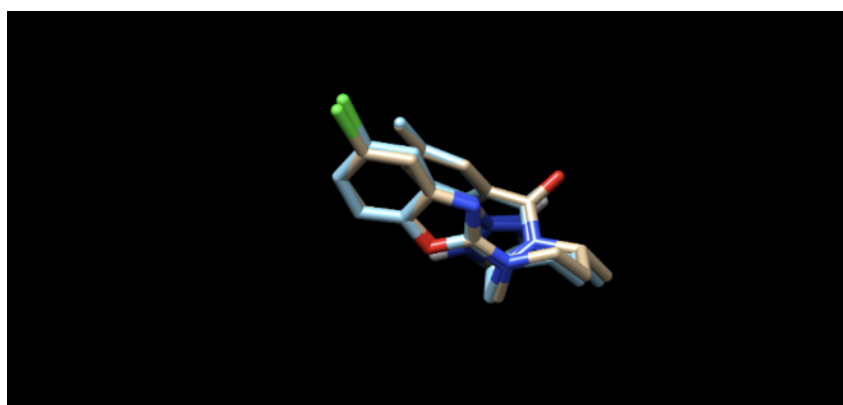


Figure 3: Redocking results: Suvorexant redocked pose (orange) vs. co-crystallized pose (blue). RMSD = 0.22 Å.

pocket and that high stability is maintained through a dense network of Van der Waals contacts and hydrophobic interactions with key residues of the receptor (Figures 4 and 5).

A summary of all docking results is presented in Figure 6.

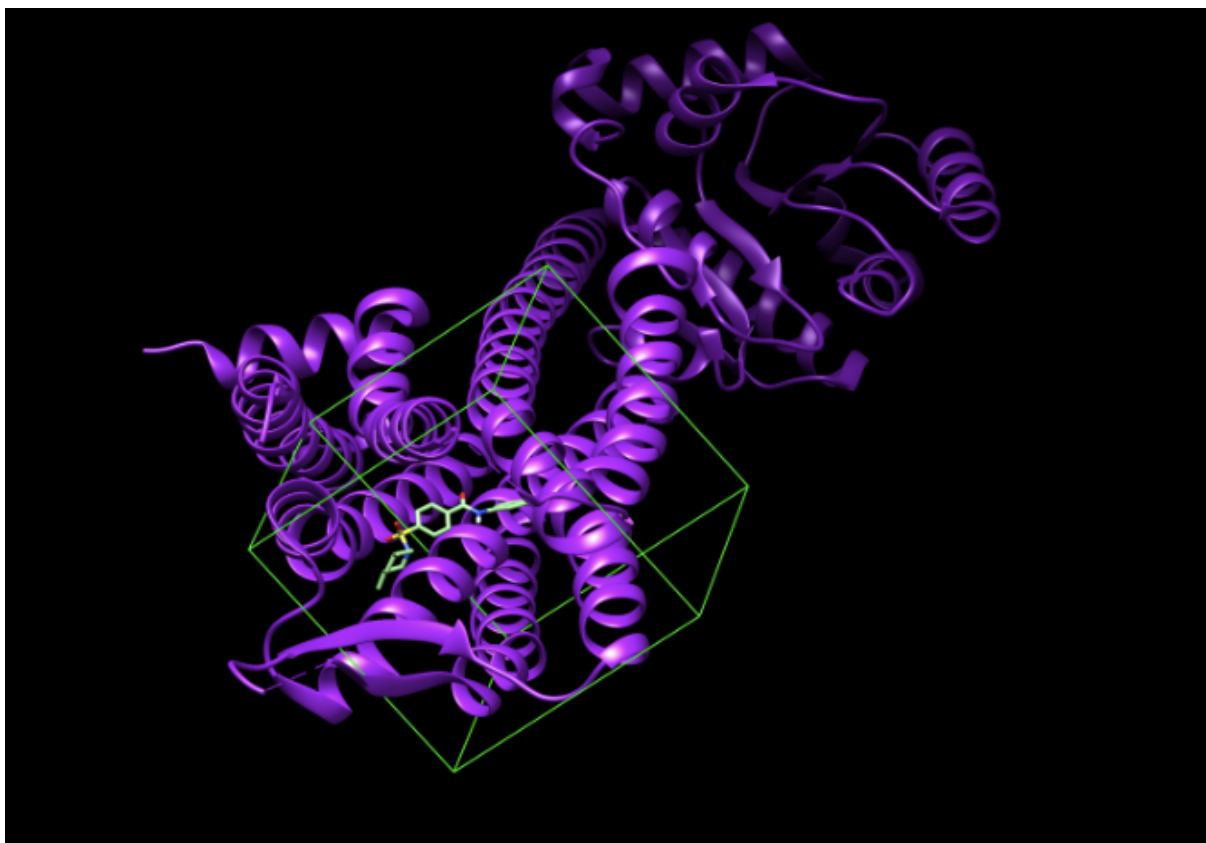


Figure 4: Z26441190 interaction with the 4S0V OX2 binding pocket.

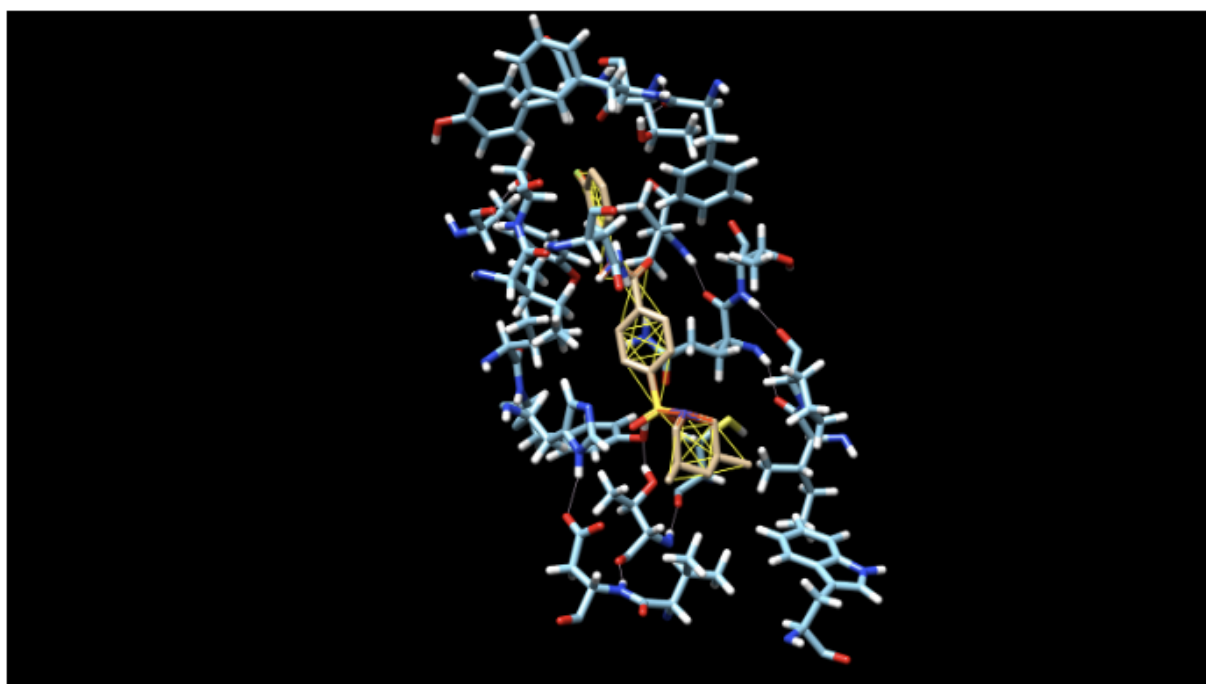


Figure 5: Z26441190 interaction analysis with the binding site of the 4S0V OX2 receptor. Yellow contacts indicate Van der Waals interactions.

4. Conclusions

The presented *in silico* research successfully implemented a multi-step virtual screening workflow to identify novel antagonists of the human Orexin 2 Receptor (OX2), a validated

Catalog ID	Compound Type	Affinity (kcal/mol)	Interpretation
Suvorexant	Reference (+)	-11.56	Benchmark
Z26441190	Screened compound	-10.49	Very Strong (Top Hit)
Z28143004	Screened compound	-9.93	Strong Affinity
Z1155102784	Screened compound	-8.98	Significant Affinity
Z52198758	Screened compound	-8.58	Strong Affinity
Z1840003394	Screened compound	-7.69	Moderate Affinity
Aspirin	Control (-)	-5.73	Weak (Non-specific)

Figure 6: Docking results for all five candidate compounds and controls.

target for the non-addictive treatment of insomnia.

The integrated pipeline was rigorously validated using controls. The QSAR model correctly predicted the potency of Suvorexant and Lemborexant, while the pharmacophore model was able to distinguish between OX2 receptor antagonists and benzodiazepines, ensuring that the identified hits present the structural and chemical requirements for inhibiting OX2.

Among the screening library, compound Z26441190 was identified as the most promising and potent lead candidate. This molecule showed an exceptional binding affinity of -10.49 kcal/mol and a high QED score of 0.851, demonstrating both favorable pharmacokinetic properties and drug-like characteristics.

The significant energy gap between the lead candidate and the negative control (Aspirin, -5.73 kcal/mol) confirms the high specificity of the binding. Furthermore, structural divergence from benzodiazepines or GABAergic modulators supports the hypothesis that these novel scaffolds provide a safer and non-addictive alternative to traditional sedatives.

Subsequent *in vitro* binding assays and functional studies for the identified lead Z26441190 will be required in order to confirm its antagonistic activity and safety profile.

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